

Oracle Beater Manual

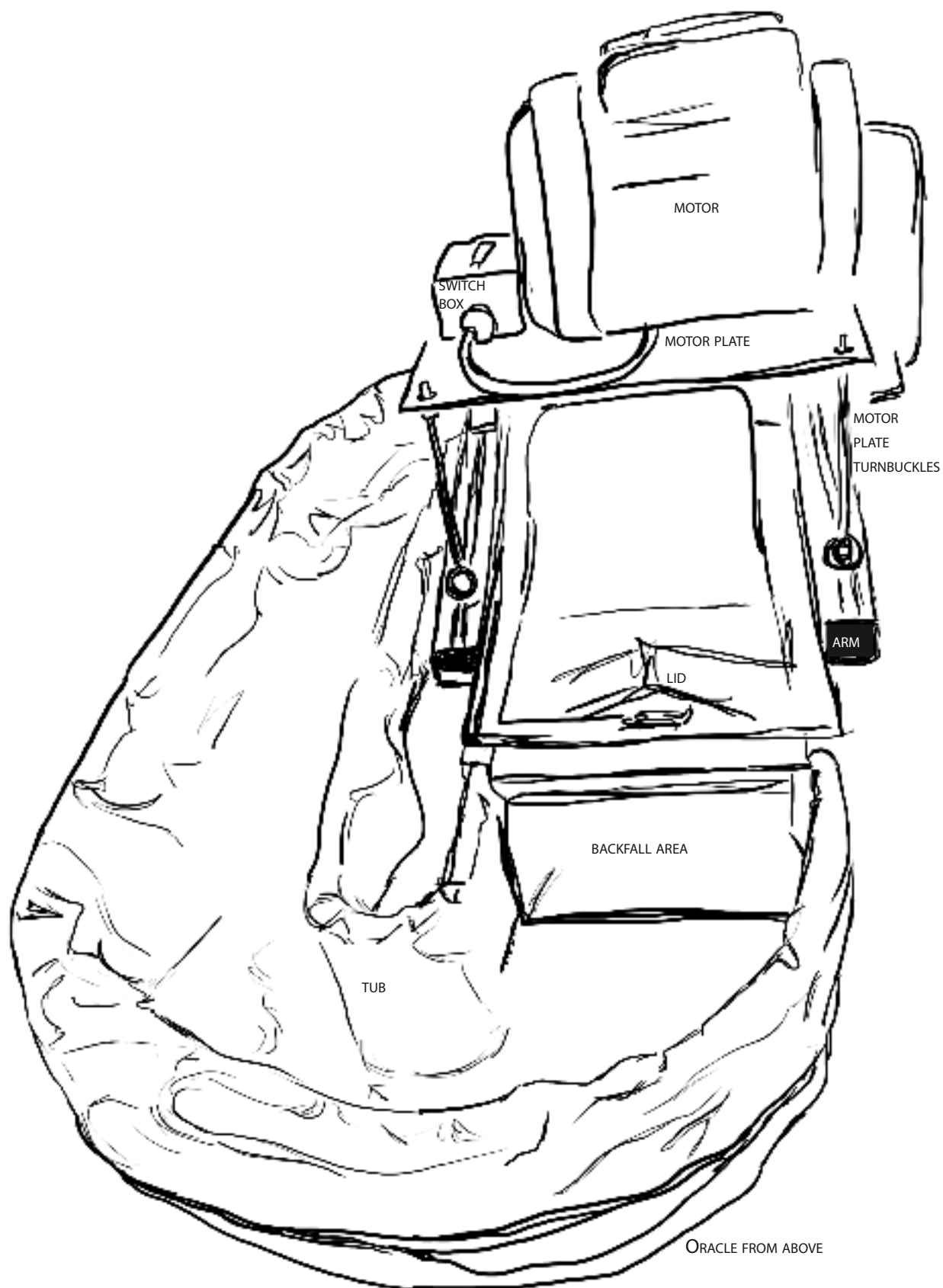
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revised 2012

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This manual is intended to help you use your beater to its full potential. Before setting up the beater, read at minimum the sections on safety, setup, and basic use. We've also provided a section on specific fibers for reference as you move on to different techniques.



Why use a beater?

A beater shortens fibers, breaking them at their weakest point by forcing material between the roll and bedplate. It also splits fibers externally and internally, creating fibrils — this is the process called fibrillation. External fibrillation also removes the primary fiber walls.

These shorter fibers can mesh to form a uniform sheet of paper. Fibrillation strengthens the final sheet by increasing the surface area available for hydrogen bonding as well as softening the fibers and making them receptive to dyes and filling agents. With a beater, you can turn any properly prepared cellulose-containing plant material into pulp for papermaking. You can also rehydrate dried pulp, changing its characteristics to suit your technique.

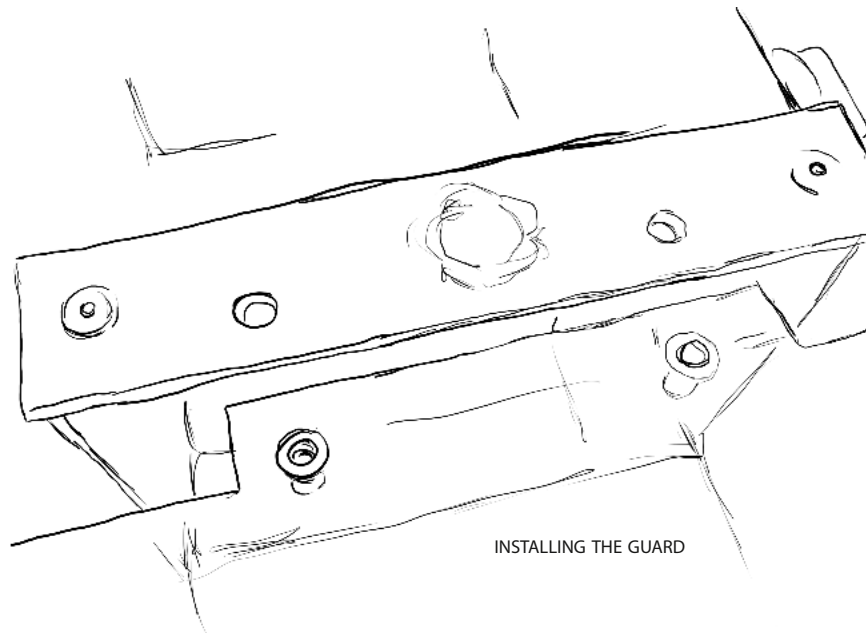
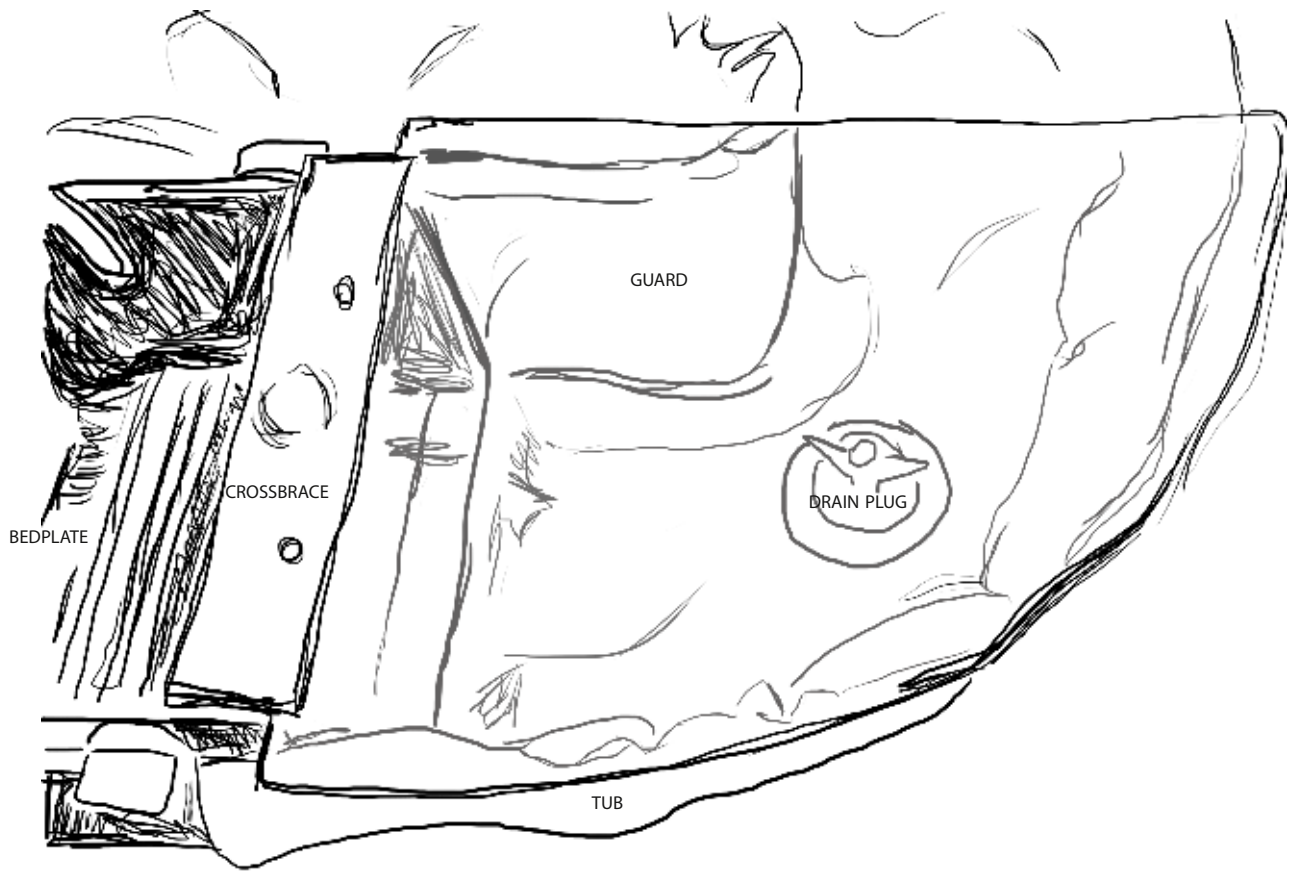
About this beater:

Capacity:

1 lb of raw fiber; 1.5 lbs of sheet pulp

Construction:

6x7 stainless steel roll and bedplate
PVA polyester fabric tub
Aluminum core
1/2 horsepower TEFC motor 110/220 volts



Safety Warnings and Information

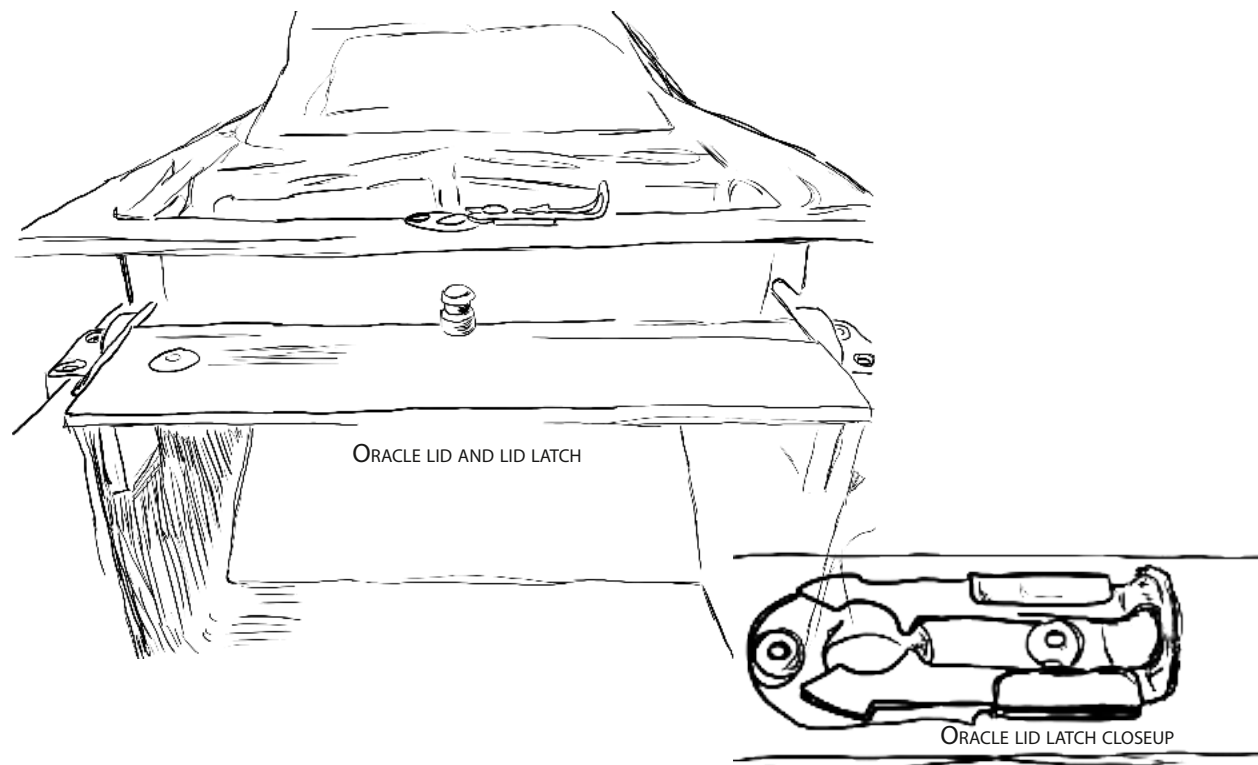
1. THE ORACLE BEATER CONTAINS ROTATING PARTS, WHICH CAN CAUSE PERSONAL INJURY. DO NOT ATTEMPT TO OPERATE THE MACHINE WITHOUT THE LID AND GUARD IN PLACE. DO NOT INSERT HANDS OR FINGERS UNDER LID OR GUARD WHILE THE MACHINE IS OPERATING. DISCONNECT POWER BEFORE CLEANING OR MOVING THIS MACHINE.

2. The manufacturer disclaims any and all responsibility unless this unit is operated in compliance with all federal and local regulations. The guards must be in place and in good operating order. Removal of any parts or accessories from this machine is in violation of current federal and local safety and health laws. By acceptance of this merchandise, the purchaser and the user assume complete responsibility for the safe operation of this equipment.

3. UNPLUG THIS MACHINE BEFORE CLEANING. DO NOT SPRAY THE MOTOR OR ANY OTHER ELECTRICAL PARTS WITH WATER! USE A SPONGE TO WIPE OFF THE SURFACES OF ELECTRICAL PARTS.

4. This machine contains plastic parts. Do not expose plastic parts to solvents or boiling water. Do not use abrasive cleaners on this machine. Clean with suitable aluminum and PVC cleaners; many marine cleaning supplies are appropriate.

5. As a rule do not add any chemicals to the pulp being beaten. Chemicals such as dyes and pigments can adhere to the sides of the tub and will discolor it.



Electrical Information

The Oracle is wired for 110 volts A.C. (60 Hz.), but can also be wired for 220 volts A.C. To wire the beater for 220 volts, follow the instructions on the motor for wiring to 220 or “high” voltage, exchanging the plug supplied for the proper 220-volt plug (waterproof). We can also supply 50 Hz motors for overseas use.

GFCIs:

We recommend that you use ground fault circuit interruptor (GFCI) outlets with the beater and other electrical equipment in the paper studio. A GFCI provides protection from electric shock hazards caused by ground faults, preventing you from being the path of least resistance for electrical current in the case of grounding failure in an electrical device. If you are adding a separate circuit to supply the beater, follow local and state electrical codes. Place new outlets high on the wall, away from splashing. Avoid using extension cords, as there is an increased risk of electric shock.

OSHA standards:

The motor control on this machine does not conform to current OSHA standards. If you require OSHA-standard equipment, please contact us for more information.

Switch lockout:

If you wish to prevent unauthorized use, padlock the switch box cover (on non-magnetic starter models) closed.

Additional useful equipment

- Floor protection

Papermaking is wet, no matter how you do it. Ideally, you will have a waterproof floor with a drain, but a shower stall base plumbed with a drain will work. You can also use heavy tarps as a temporary solution.

- Buckets

A 20-gallon trash bucket fits under the beater and holds a whole load. A collection of several 5-gallon buckets is also handy.

- Water supply

You will need a way to fill and clean the beater, and a hose with a spray nozzle is best. A detergent spray bottle (not a trigger spray bottle) or buckets of water will also work.

- Sponges

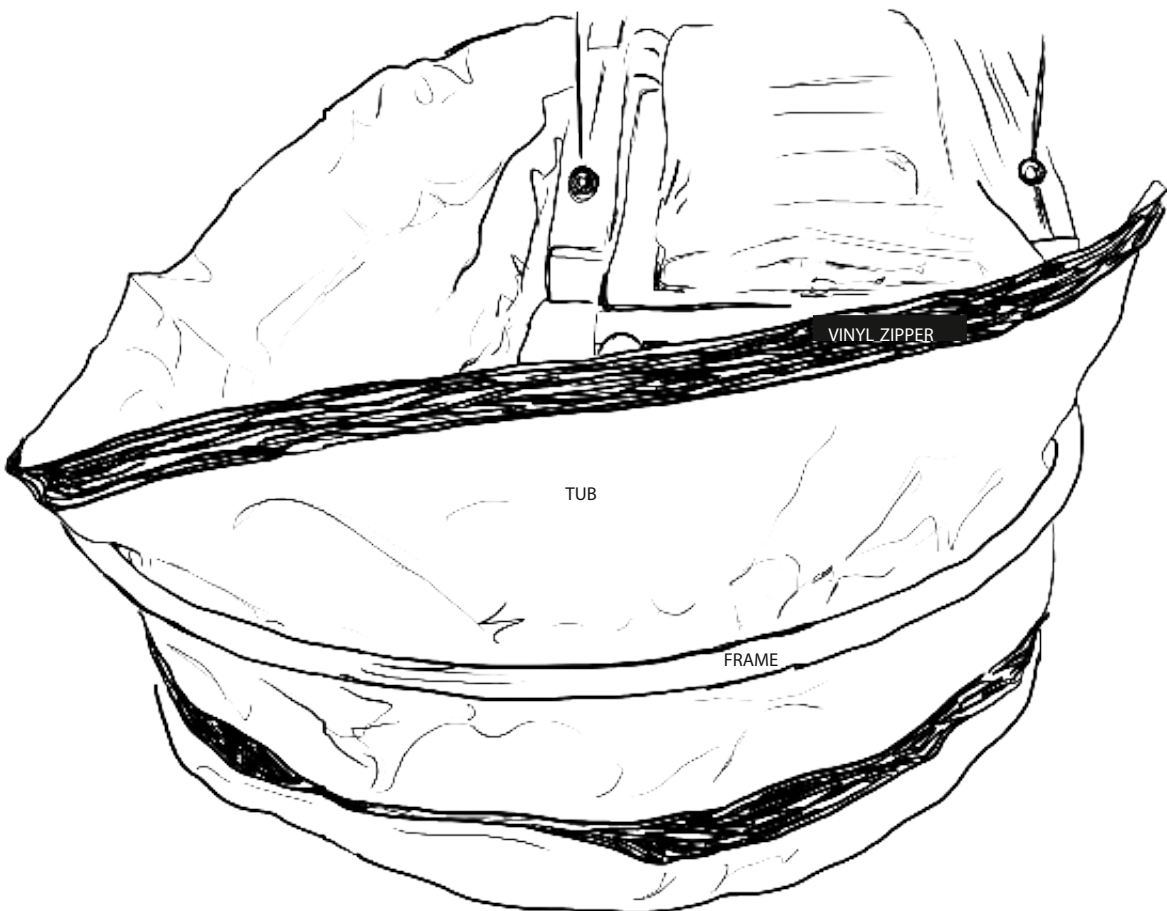
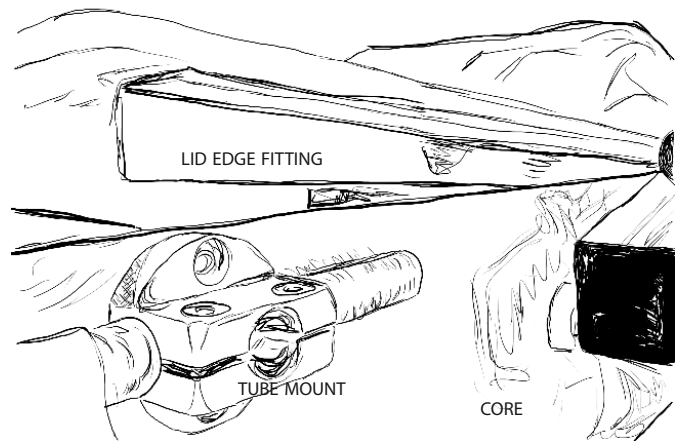
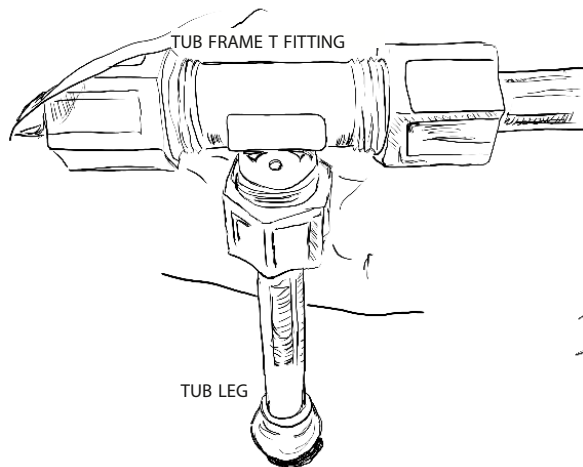
You'll want several big ones, large enough to span the return side of the tub.

- Hearing protection

Beaters are loud, so if you're going to be in an enclosed space with it for extended periods of time, earmuffs or earplugs are a must.

- Strainers

Having something to strain leftover pulp during cleaning keeps your drains from getting clogged.



ORACLE WITH TUB UNFOLDED OVER FRAME

Assembly

PARTS

- 1 Beater assembly (core)
- 4 plastic feet (core)
- 2 small curved tubes (frame)
- 2 large curved tubes (frame)
- 2 straight tubes (frame)
- 4 legs with T fittings (frame)
- 2 M8 washers (guard)
- 2 M8 lock washers (guard)
- 2 M8 hex bolts (guard)
- 2 allen wrenches
- 1 tub guard
- 1 drain plug
- 1 beater manual

STEPS

1. Unpack the beater. CAUTION: It is bolted to a supporting pallet, and it is heavy. Lift the whole core, not just the arm. Set lid and guard (wrapped separately) aside temporarily.
2. Unbolt the beater core from its pallet.
3. Attach beater core feet to threaded rivets on underside of core where it was attached to the shipping pallet.
4. Assemble the tub.
 - a. unfold the tub
 - b. attach straight tubes to legs
 - c. insert longer end of large curve into tube mount on outside of core
 - d. insert longer end of short curve into tube mount on outside of core
 - e. attach legs to curved pieces
 - f. check tub for fit by making sure it lies mostly flat inside frame and sides are vertical; adjust curved frame pieces as necessary
 - g. tighten all plastic fittings and hex bolts in core tube mounts (using included allen wrench)
 - h. fold tub over frame and fasten black vinyl zipper
5. Unpack motor and mount plate assembly.
6. Attach the motor plate to the large brackets on the arm crossbrace with M8 bolts. Insert the bolts from the top, and loosely tighten nuts. The bracket pairs are slightly adjustable, so move them if necessary to get everything to line up correctly.
7. Make sure the belt guard back pieces are correctly aligned with each other and attached.
8. Position the turnbuckles at the backfall side and insert the bolts (which are packed inserted into the turnbuckle) into the small bracket on the motor mount plate. There should be a washer on both sides of the bracket, and a nut with a nylon insert to lock everything in place. Refer to diagrams on the following page as necessary.
9. Tighten all nuts and bolts down completely.
10. Slip V-belt onto the motor pullies.

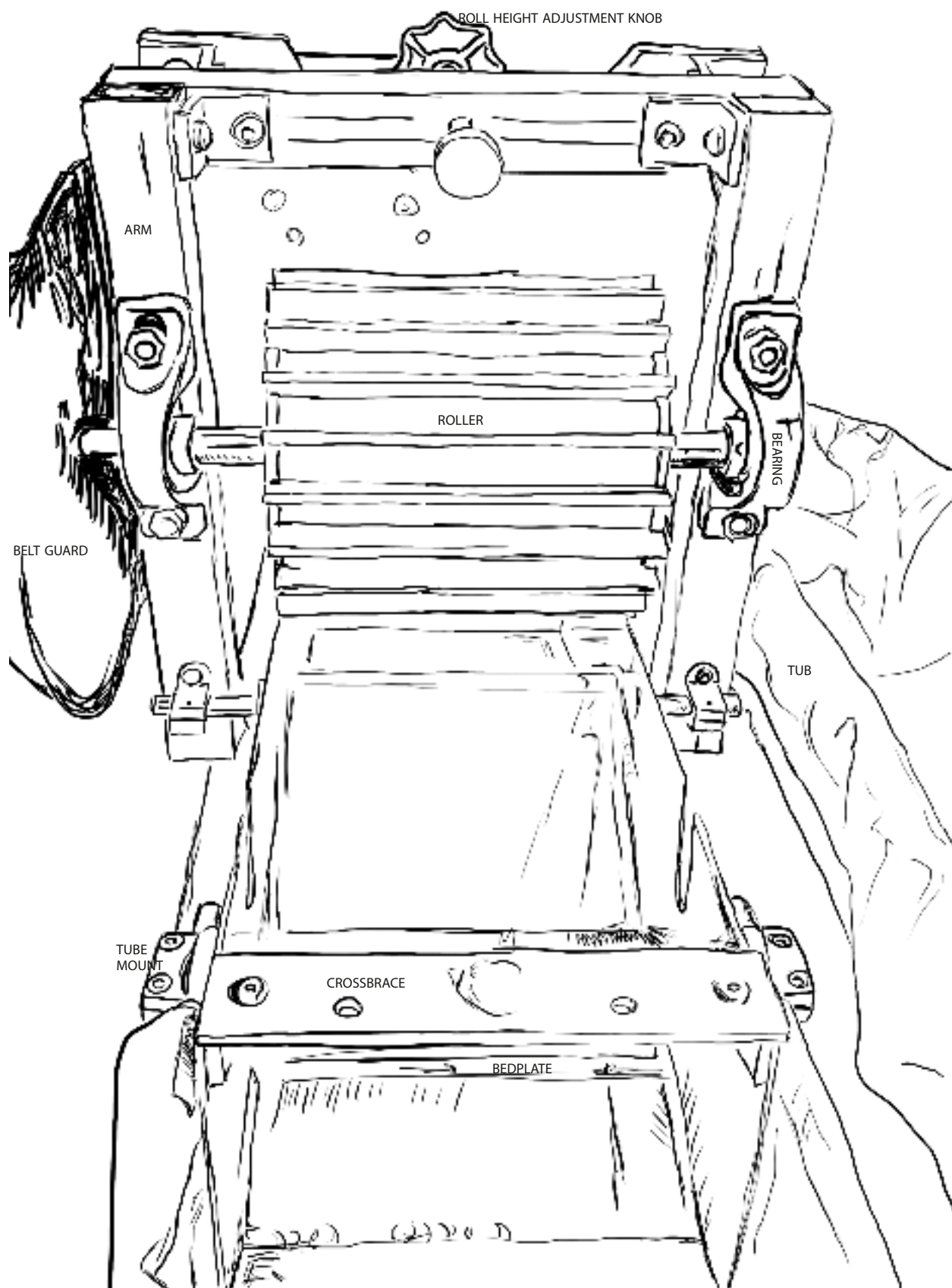
11. Adjust the turnbuckles by loosening the nut and turning the center pieces. Adjust them in tandem until the motor mount plate is level and there is a half-inch of deflection on the belt. Tighten the nuts back down with a wrench to prevent the turnbuckles from loosening unexpectedly.
13. Attach the outside of the belt guard (black plastic) with included M8 bolts. There are two bolts with one washer and lockwasher each that go in the holes aligned with the arm crossbrace, and one bolt in the top hole on the guard that has two washers (one for each side of the plastic) and a nut with nylon insert.
12. Unpack and attach splash guard to intake-end crossbrace with included bolts, lockwashers, and washers.
13. Make sure the roll adjustment screw and wing nut work smoothly. Adjust the roller so you can spin it by hand and fit the lid.
14. Check to be certain that there are no loose parts, such as washers or bolts, in the tub.
15. Unwrap the lid and fit it over the roll. Lock the lid latch & engage other safety features if necessary. Plug the machine in, turn it on, and check that the motor comes on and turns the roll.
16. Pre-clean the beater to make certain that any debris from construction and shipping is cleared. Fill it with water and run for five minutes, then drain and wipe down the tub.
17. Make paper!

turnbuckles!

The Quick-Start Checklist for Beating

1. Prepare fiber.
2. MAKE SURE THE BEATER IS UNPLUGGED.
3. Check tub for foreign objects.
4. Check that the roll is NOT in contact with bedplate.
5. Replace drain plug.
6. Make sure the lid is on.
7. PLUG IN.
8. Turn on.
9. Add water.
10. Add fiber.
11. Clear pulp with roll raised.
12. Engage roll and beat as long as you like.
13. Raise roll and clear pulp for ~5 minutes; add internal sizing now.
14. Empty pulp from tub with the machine running.
15. Turn off AND UNPLUG beater.
16. Remove lid and clear remaining pulp with hose or sprayer.
17. Clean beater.

Based on the quick-start guide written by one of our customers



Step-by-step Beating Instructions

1. Prepare fiber

¶ Measure out sufficient material for a full beater load. This is 1.5 lbs of sheet pulp or 1 lb of rag or raw fiber. In certain cases, you will be able to process slightly more and in others you'll be able to do less. Experiment to determine how much of specific fibers the beater can handle to your satisfaction.

¶ To avoid clogging the beater, cut fabric and raw fibers to pieces around 1" or smaller.

¶ Sheet pulp and prepared pulps do not require advance preparation.

¶ Raw materials, such as plant matter and rag, need to be soaked. If possible, soak your material overnight. This allows the water to fully penetrate the fiber, fully saturating it and making the early stages of beating easier. Some fibers will work without pre-soaking, but you will have to experiment to determine which. Make sure the fiber is fully submerged when soaking, covering it with a board if necessary.

¶ Some stuff requires cooking in a caustic solution before beating. CAUTION: Wear gloves and use adequate ventilation when working with caustic material. Usually, the caustic is soda ash. The specific proportion of soda ash to water depends on the fiber you're cooking and how much you wish to break it down chemically. Be sure to rinse the cooked fiber completely before you put it in the beater.

2. MAKE SURE BEATER IS UNPLUGGED.

3. Check tub for foreign objects.

¶ Always make certain the beater is unplugged when you're checking it over before beginning a beat. Check the tub for foreign material that can contaminate your pulp or damage the roll before you begin. NEVER PUT YOUR HANDS IN THE ROLL AREA IF THE BEATER IS PLUGGED IN.

4. Check that the roll is NOT in contact with bedplate.

¶ Make sure that the roll is NOT in contact with the bedplate before beginning. Spin it by hand to check, and use the adjusting screw to raise it until it's barely touching the bedplate. Then raise it four additional turns for clearance.

5. Insert drain plug.

6. Make sure the lid is on.

¶ Put the drain plug in from the bottom of the tub, and make sure it's screwed tight. Position the lid so its metal parts are inside the tub and it has hit the backstop. It should fit snugly inside the core. Lock the lid latch.

7. PLUG IN.

8. Turn on.

¶ Double check to make sure the lid is on, safety features are engaged, and the drain is closed. Then plug the machine in and turn it on.

9. Add water.

¶ Sometimes you may want to pre-clean the beater, especially if you haven't used it for a while or you're unpacking it after shipping. Simply fill the tub completely and run it for five minutes with a load of plain water. This will clear out any leftover fiber from previous use. Empty and refill the beater before starting your fiber.

¶ Use a hose (or bucket) to fill the beater to a couple of inches below the lip of the tub. You want to start with enough water so it circulates over the backfall.

10. Add fiber.

¶ Always begin with small pieces of pre-soaked material to prevent clogs (except in the case of sheet pulp, which doesn't require soaking). Watch for returning material in the opening so you know it's circulating. Once it has started to flow properly, you can add slightly larger pieces if you wish. Always add fiber close to the backfall in order to allow it to soak up water and float freely.

¶ If you can't get the measured pulp into the beater before it stops circulating, try adding more water. Watch the backfall, though, as too much water may cause it to back up.

11. Clear pulp with roll raised.

¶ Begin by leaving the roll in its raised position for up to five minutes — the amount of time depends on how lumpy your stuff is. This will even out the pulp flow and reduce clumps. At this stage, you will probably see the whole arm jumping. This is normal: it's designed to move to avoid damage from large clumps, and to allow harder beating at the beginning of the process.

12. Engage roll and beat as long as you like.

¶ After the flow of fiber has evened out, lower the roll until you hear it starting to grind. The motor gets louder as it works harder, and in general beaters make a lot of noise. Don't be too concerned about it, but do wear hearing protection. If you overwork the motor, the thermal protection is designed to shut it off before it sustains damage. If you trip the motor reset or the circuit breaker, allow the motor to cool down before attempting to reset it. You may need to add some water to restore the pulp flow. If the thermal protection trips often, it's possible you're beating too hard.

¶ Adjusting the roll position to achieve the desired degree of beating action is like driving a manual transmission car without a tachometer: you can judge how hard you're beating the fiber by the amount of noise the beater is making and the quality of the sound it makes. The sound will even out at a given setting as the fiber is processed. For hard beating, lower the roll until the grinding is very loud and the motor seems to be straining, then raise it a half-turn or so.

¶ Initially, you will need to experiment to get a sense of how hard you can work the beater.

¶ Always remember to tighten the wing nut on the adjusting screw to hold the roll in position.

¶ When the sound of pulp processing evens out at a given setting, lower the roll to beat the material further.

¶ Rag breaks up slowly and will require careful attention, while sheet pulp tends to be faster and can be left alone more safely.

¶ If you prefer a less idiosyncratic method of gauging beating action, you can add an amp meter. This allows accurate setting of the roll based on power draw. There are several options for doing this, including watt meters (you can also try searching for electricity monitors), which are readily available at home improvement stores.

Test the pulp

¶ During the beat, you'll want to check the progress of the pulp. The best way to do this is to scoop a finger-full of pulp from the return, put it in a clear jar with some water, and shake the jar. This doesn't require stopping the beater. Look at the pulp in the jar against a light and observe key characteristics of the pulp:

- Presence of clumps
- Fiber length
- Knots

¶ What you see allows you to determine whether or not the pulp is done to your satisfaction.

¶ Squeezing pulp through your fingers also reveals a lot about its quality. A long beating time, for example, makes a pulp that feels greasy. Don't put squeezed pulp back in the beater, as it can create knots.

¶ Experience will give you the ability to judge pulp characteristics, so practice as much as possible. Pay close attention throughout the beating process to familiarize yourself with the way fibers change as they're processed.

13. Raise roll and clear pulp for ~5 minutes; add internal sizing now.

¶ Raising the roll at the end "brushes" the fiber, which removes knots that form during beating. A load should be cleared for at least two minutes, or as long as five if you are particularly concerned about knots. This is also the time to add internal sizing, so that it gets completely absorbed into the pulp.

14. Empty pulp from tub with the machine running.

¶ The pulp should be drained while the machine is still running, so all the material moves out of the roll area. Before draining, be sure that the roll is completely disengaged from the bedplate. Put a bucket under the drain. Reach under the tub and remove the drain plug. Let the pulp drain, and turn the machine off once the pulp has stopped circulating.

15. Turn off AND UNPLUG beater.

¶ CAUTION: Never open the beater or clean the roll without unplugging it first.

16. Remove lid and clear remaining pulp with hose or sprayer.

¶ Clear out any remaining pulp with a hose. If you're going to switch to a different pulp, rinse the beater out completely. If you're continuing with another load of the same material, it isn't necessary to get all the pulp out.

¶ Remember to replace the drain plug before refilling the beater.

17. Clean beater.

CLEAN THE BEATER AFTER EVERY DAY OF USE.

It is necessary to completely clean the beater at the end of every day of use. Pulp that is allowed to dry is difficult to remove and can contaminate future batches.

1. Fill the beater with clean water and run it for five minutes.
2. Drain it.
3. Turn the machine off and unplug it. ALWAYS UNPLUG IT BEFORE CLEANING THE ROLL.
4. Remove the lid, rinse any pulp off it, and set it aside to dry.
5. Raise the arm and clean the roll. CAUTION: The arm and motor assembly weighs about 35 lbs. USE CARE WHEN RAISING AND LOWERING THE ARM. Dropping it can cause injury or damage to the machine. Pay special attention when cleaning the roll and bedplate. Use a hose to flush out any pulp remaining between the bars, and check for pulp on the shaft. Be sure to also completely flush out the bedplate.
6. Clean the tub with water and a sponge. DO NOT use solvents, abrasives, or boiling water on the tub or other plastic parts.
7. Clean the motor and motor plate by wiping it with a sponge. Always check to make certain the machine is unplugged. DO NOT SPRAY THE MOTOR OR OTHER ELECTRICAL PARTS WITH WATER.
8. Wipe everything dry with a clean sponge.

Troubleshooting

Beater does not run

¶ If the beater does not run when switched on, first check that the beater is plugged in and the outlet has power.

¶ Next, check that the motor's thermal protection has not been tripped. If it has, wait until the motor is sufficiently cool to reset the protection. If the thermal protector has tripped, you have to wait until the motor cools down enough to allow the protector to reset. The motor does not reset automatically. You must push the reset button on the motor, and there will be an audible click if it resets correctly. Otherwise, check the circuit breaker in the wall (and the GFCI if installed), to see if they have been tripped.

¶ If the roll is in contact with the bedplate when you start the beater, the motor will work but the roll won't turn. Raise the roll so it no longer touches the bedplate and restart the beater.

Beater clogs when material is loaded

This will happen if:

- the fiber is not properly prepared
- the pieces being loaded are too large
- it is loaded too quickly
- the water to fiber ratio is too low

¶ First, try filling the beater with more water. This will solve many clogging issues.

¶ If this is unsuccessful, turn the machine off, remove the lid, and raise the roll. There may be a clump between the backfall and the roll. If there is, remove some of the excess fiber and place it in a bucket temporarily. Lower the roll, replace the lid, and turn the machine back on. If the pulp is flowing again, you may begin to add the material from the bucket back in. If not, try adding more water. Once it's flowing, add the excess material back in. Sometimes it will not be possible to replace it all, so pay close attention and make sure to maintain circulation.

Beater clogs during beating

¶ If the material has been circulating properly, but stops mid-beat, the ratio of fiber to water is too high. It will be necessary to add water or remove some fiber. Usually, adding water will restore circulation. If the beater is too full, remove some pulp.

¶ Be careful, however, not to add too much water: this can also hinder circulation, as pulp will remain in the backfall area, causing a slow-down.

¶ The consistency (fiber/water ratio) of pulp that will work varies with the material being beaten. For example, this ratio can be higher when using prepared pulp than with rag. When beating raw fibers and rag, there is usually a point in the cycle where the fibers start to rope together and may stop circulating if the beater is overfilled.

Reference for Specific Fibers

The following information is designed to help in proper fiber preparation and beating when using the Oracle beater. The beating times listed are approximate and given as guidelines only. They will give you points of reference for making your own papers.

RAG—COTTON, LINEN, OR OTHER NATURAL PLANT FIBER

1. Preparation

Rags must be cut into small pieces, about 1" x 1," before being beaten. You can use slightly larger pieces if you are careful and patient when loading the beater. Make sure the rags are clean and free of zippers and seams. Synthetic blends will not beat properly. They will leave threads and tend to form strings in the beater. All rags should be rinsed well to remove soaps and detergent or they will foam when beaten. Rags should be pre-soaked overnight or cooked.

2. Beating

Set the roll in its raised position to establish pulp flow, and then lower the roll to a position where the roll assembly starts to bounce a bit but the motor doesn't stall. If the motor does stall, lift up on the arm to free the roll. If this doesn't work, turn the machine off and take care of the clog. Once the beater settles down and isn't jumping too much, lower the roll more. After the rags have been broken up, you can settle into a beating cycle, and the load will not have to be watched as carefully. Depending upon the toughness of the fiber, beating time may run from 1/2 hour to several hours. Run the beater in the raised clearing position for 5 minutes.

3. Notes

In order to soften rag fibers and make them easier to beat, sometimes they are cooked before beating. This is usually done with a caustic solution such as soda ash. The machine will bounce as it starts to break up rag. The beater can handle 1/3 less weight of rag per load than cotton linters (see capacity specification). Watch the pulp flow as the rag begins to break down and increase the viscosity of the pulp. The viscosity may get high enough to stop the pulp flow. At that point, you will need to add more water to get the pulp flowing again.

RAW FLAX

1. Preparation

Flax should be soaked for at least an hour, though it is best to soak the fibers overnight. After soaking, the flax fibers should be thoroughly rinsed in running water. Cleaning will help to reduce foaming during beating, although it may not prevent it. Flax fibers may be cooked in a caustic solution prior to beating, as with cotton rag. The fibers should be cut into 1" — 2" pieces before loading them in the beater.

2. Beating

Set the roll in a raised position, and add the fibers slowly. Lower the roll and wait until pulp flow evens out. Lower the roll again so that you are getting a medium to hard beating action. Beating time is 1 – 10 hours. Run the beater in the raised clearing position for 5 minutes.

3. Notes

The longer the fibers are beaten, the more translucent the paper made with them will be. When beating flax for a long period of time, use a lighter beater setting.

OTHER RAW FIBERS

1. Preparation

Raw natural fibers need to be cleaned and cut into small — 1" long — pieces. Longer pieces may work, but there is a danger of clogging. For the best results, cooking in a caustic solution before beating is usually required. Soak the fibers overnight if possible to make sure that they are sufficiently waterlogged to sink and go through the machine. Some varieties of fiber (e.g. cattails) will almost always float. This makes these fibers almost impossible to beat by machine. You should be able to beat these fibers by hand or with a stamper.

2. Beating

Beating procedures for raw fibers vary widely, depending on the fiber used. Some fibers will require little or no beating and can be made into pulp with the roll in a raised position during the whole cycle. Others will require very heavy beating to break them down. Run the beater in the raised clearing position for 5 minutes.

3. Notes

Experiment with mixing fibers for strength. With raw fibers in particular, pay close attention to how well it circulates; some raw fiber will expand dramatically mid-beat. If this happens, just remove some of the excess fiber and continue, adding the extra back in only as far as the beater can handle without backing up again.

COTTON LINTERS — FIRST OR SECOND CUT

1. Preparation

You can add the sheet pulp directly to the beater without pre-soaking. Wetting the linters will make them easier to tear. Tear them into pieces no bigger than 4" square. Too large pieces will clog, and you run the risk of finding unbeaten sheets in the bucket after beating. You can soak them overnight or mix them with a whiz mixer if you are so inclined.

2. Beating

Start with the roll in the raised position until the lumps are gone — until the arm assembly stops jumping. Beat at a medium-heavy setting for 15 minutes, or a lighter setting for up to 60. Cotton linters should be cleared for 5 minutes.

3. Notes

Cotton linters are one of the most widely used pulps. They can be used for some sculptural work by simply rehydrating the pulp, which takes about 5 minutes — to break the sheet of linters apart. By beating cotton linters for a longer period, paper strength can be increased greatly. Since cotton linters tend to have short fibers, experiment with softer beating in order not to shorten them too much.

ABACA

1. Preparation

Presoak for at least 15 minutes (ideally overnight) so they will tear easier. Tear into small pieces. Larger pieces can be added as the pulp begins to circulate.

2. Beating

Beat abaca pulps with the roll in a raised position for 5 minutes. Lower the roll and beat the abaca hard for a short time (15 minutes), or beat the abaca longer at a lighter setting (1 – 3 hours). Clear the fiber for 5 minutes.

3. Notes

Abaca pulp can be used to make paper of widely varying characteristics, from thin tissues to thick heavy papers. This is true because it can be beaten in many ways and because it is composed of long fibers. Abaca pulp mixed with cotton will increase the strength of the mixture. Longer beating time for abaca pulp will result in very translucent papers, though they shrink a lot and tend to be hard to dry.

Keeping Records

In order to obtain reproducible results, you will find it necessary to keep a record of beating times and roll positions for various fibers. Keep track of the details of every beat, including how many turns of the adjusting knob the roll is above the bedplate. We have provided a sample beater record sheet as a starting point. Feel free to copy it for your own personal use. (see following page)

Date	Fiber Type		Fiber Preparation	
	Dry Weight		Water (gallons)	
	Start Time	End Time	Elapsed Time	Notes
Loading				
BeaterSetting1				
BeaterSetting2				
BeaterSetting3				
Clearing				
Total Time				
Notes: & Sheet Sample				

Basic Maintenance

Because the wet environment and constant vibration during use is hard on some components of the beater, regular preventative maintenance is necessary to keep your Oracle beater in the best operating condition.

1. Clean the beater after every use. DO NOT USE ABRASIVES OR SOLVENTS ON PLASTIC PARTS.
2. Lubricate the roll bearings regularly: once a week with light use, daily with heavy use. Use #2 consistency Lithium-base grease formulated from a high quality mineral oil with rust and oxidation inhibitors and a grease gun with standard fittings. The grease nipples on the bearings are accessible from underneath the arms when they're in the upright position. Use enough grease that a little comes out of the bearing housing. Clean up any excess so it doesn't get into the beater.
3. Check belt tension and alignment periodically. If the belt is loose, the power transferred from the motor to the roll will be reduced. Remove the belt guard. Tighten the belt by adjusting the turnbuckles attached to the motor plate, starting with the one closer to the belt. Loosen the locking nut on the turnbuckle before attempting to move it. Check the tension as you proceed. There should be 0.5" of deflection when you push the belt. The motor mount plate should be level, so adjust both turnbuckles as necessary. Tighten the locking nuts on the turnbuckles to hold the motor plate in position and replace the belt guard.
4. Check all nuts and bolts periodically, tightening them when necessary.
5. Check all electrical connections for tightness and leaks. Tighten or replace them when necessary.
6. Arm pivot points are constructed so they do not require lubrication.
7. If the tub sustains minor damage, patch it with similar material and a waterproof vinyl adhesive. Contact us if you'd like a small piece of the original material.

Roll and Bedplate Touchup

Over time, the action of fibers between the roll and bedplate dulls the blades. This results in inconsistencies in processing and longer beating times. The roll can also get out of round, which results in uneven beating. To restore these surfaces to their original sharpness, it is necessary to touch up the roll with a load of pulp and 80-grit carborundum powder. Any time you remove the roll from the beater, it is necessary to repeat this procedure in order to restore the relationship between the bedplate and roll.

Perform basic maintenance on the machine in advance: tighten or replace belts, lubricate bearings and replace them if necessary.

1. Inspect the roll to check the starting appearance of the blades. Use a file to remove any burrs on the bar edges before you start to grind it in.

2. Start a load of low grade pulp, such as 2nd cut cotton linters. Treat it as any load of pulp. Evenly sprinkle 3/4 cup of carborundum powder on it as soon as it is circulating smoothly.

3. Once the carborundum powder is dispersed, slowly lower the roll until you hear a grinding sound. It will be quieter than the sound of full beating. Leave it at this point until the sound diminishes. At that point, lower the roll until it gets loud again. Lower the roll in small increments and let it grind in for a while before lowering the roll again. Do not rush. Check the beater every 10-15 minutes. Staying too long at a given point won't hurt the beater, and lowering the roll too fast can be damaging — it can make an out of round roll even more out of round.

When the roll is ground in properly, the sound it makes will be an even growl that changes in volume without changing in quality as the roll is lowered. The process takes about an hour, unless the roll is out of round or nicked, in which case it will take longer.

At this stage, check to see if all the surfaces of the bars have been sharpened. They will appear to have a different shine than they did when you started.

Lower the roll the as few times as you can, while still sharpening all the bars.

Remember that sound is subjective, and a beater can make more noise than is comfortable without ear protection and still not be at full beating intensity. The idea is not to be too gentle, while not being aggressive.

If you have had to grind in a lot, there may be nibs appearing on the outside of the bedplate. These should be ground off using a Dremel grinder.

If a ticking sound occurs when you first lower the roll, exercise caution: this means that the roll has gotten out of round and is hitting the bedplate in only one spot. Proceed with grinding in normally, but be even more patient than usual. It is vital to complete the grinding in at a particular level before lowering it, in order to re-round the roll. You will be able to set the roll location by the sound you hear. Stop lowering the roll when you hear the sound of the roll hitting unevenly. Grind it in completely, then lower it again. Do this as many times as necessary.

If, by some mischance, foreign objects go through the beater and cause deep nicks in the roll that can't be touched up by simple grinding in, they can be repaired. Fill the spots with stainless steel welding rods for 304 stainless steel and grind them nearly flush, then follow the grinding in procedure as above.

Papermaking References

This list is not exhaustive, but it will give you a number of starting points for further research. Most books do not give much information on beating, but we have included a few that provide some.

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